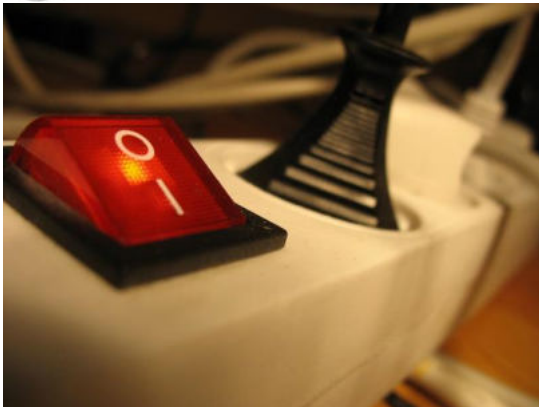


3 Data Representation Methods in the Computer System

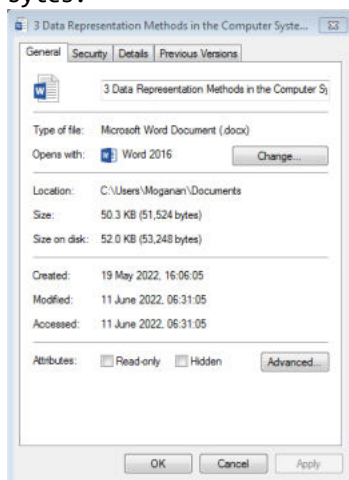
27 June 2022

Notes

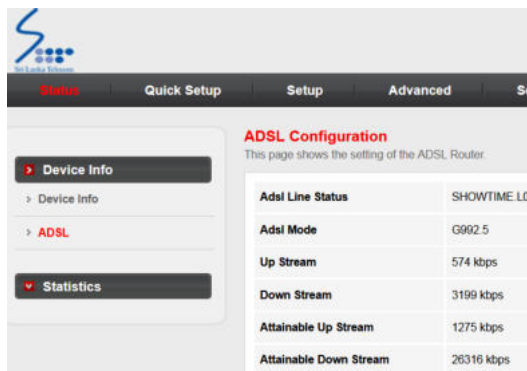
1. The switches that we come across everyday life generally have two states namely off (0) and on (1). There is no other state other than on or off. Similarly bits do have 2 states only namely 0 and 1. Off is like bulb is off and on is like bulb is on. Computer/Laptop button is used to power on and the same can be used to power down the computers (depending on the setting) long press (in case of computer stuck only, generally shutdown feature is used) can power off the computer.



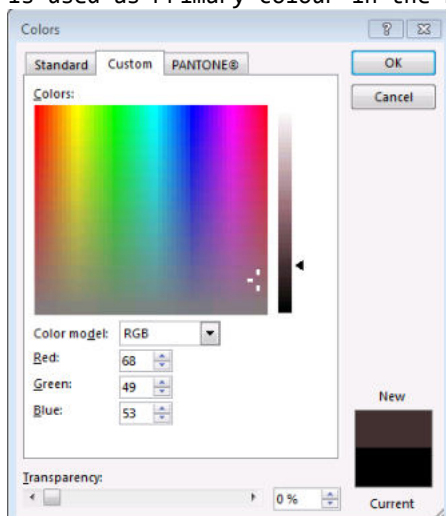
2. In the number systems we have number one less than the base i.e. decimal number system consists of numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 (Base 10 from 0 to 9).
Similarly binary number system consists of numbers 0, 1 (Base 2 from 0 to 1)
Similarly octal number system consists of numbers 0, 1, 2, 3, 4, 5, 6, 7 (Base 8 from 0 to 7)
Similarly hexadecimal number system consists of numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F (here $A_{16}=10_{10}$, $B_{16}=11_{10}$, $C_{16}=12_{10}$, $D_{16}=13_{10}$, $E_{16}=14_{10}$, $F_{16}=15_{10}$).
It's essential to mention the base of the number system.
3. Zero rule - 0th power of any number is equal to 1 i.e. $a^0 = 1$, $2^0=1$, $8^0=1$, $10^0=1$, $16^0=1$
4. One rule - 1st power of any number is equal to that number itself i.e. $a^1 = a$, $2^1=2$, $8^1=8$, $10^1=10$, $16^1=16$
5. Computer stores the data on bits however it shows the hexadecimal numbers so that it can be read.
6. Computer can process the data in bit/bits (binary digit) and saves the data in bytes.



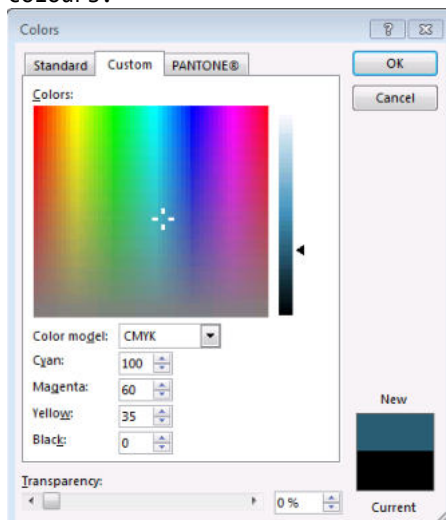
7. Bbps (bits per second) is used data communication. For example SLT (Sri Lanka Telecom) ADSL (Asymmetric Digital Subscriber line - Asymmetric means here download and upload are in different speeds) connection down speed (download) is 4Mbps (Megabits per second), upload speed is 512Kbps (Kilobits per second). Below picture shows an SLT ADSL connection speeds.



8. Computer data are stored in Auxiliary storage devices like Harddisk, SSD - Solid State Disk, Pendrive/flash drive are copied to the main memory (primary memory) then the CPU makes use of the data.
9. During the developments of monitors RGB - Red, Green, Blue were used form the other colours. Which is known the as the primary colours of the display mainly due to the fact Red Green and Blue phosphorous used in CRT monitors (Primary colours in Art subject is different from this) was used in the past. Even today same colour coding is used as Primary colour in the LED/LCD monitors.



Colour printers mainly uses CMYK (Cyan, Magenta, Yellow, Black) to form all the colours.



10. 1 Byte = 8 bit
- 1 KB (Kilo Byte) = 2^{10} Byte = 1024 Byte $\approx$ 1000 Byte $\approx$ 10^3 Byte
- 1 MB (Mega Byte) = 2^{10} KB = 1024 KB $\approx$ 1000 KB $\approx$ 10^3 KB
- 1 GB (Giga Byte) = 2^{10} MB = 1024 MB $\approx$ 1000 MB $\approx$ 10^3 MB
- 1 TB (Terra Byte) = 2^{10} GB = 1024 GB $\approx$ 1000 GB $\approx$ 10^3 GB
11. 1 MB (Mega Byte) = 2^{10} KB = $2^{10} * 2^{10}$ Byte (As 1KB = 2^{10} Byte) = 2^{20} Byte (In multiplication if the bases are same we can add the exponent)
- 1 GB (Giga Byte) = 2^{10} MB = $2^{10} * 2^{10}$ KB = $2^{20} * 2^{10}$ Byte = 2^{30} Byte
- 1 TB (Terra Byte) = 2^{10} GB = $2^{10} * 2^{10}$ MB = $2^{20} * 2^{10}$ KB = 2^{30} KB = $2^{30} * 2^{10}$ Byte = 2^{40} Byte

12. $2 \text{ MB} = 2 * 2^{10} \text{ KB} = 2^1 * 2^{10} \text{ KB} = 2^{11} \text{ KB}$
 $4 \text{ MB} = 4 * 2^{10} \text{ KB} = 2^2 * 2^{10} \text{ KB} = 2^{12} \text{ KB}$
 $0.5 \text{ MB} = \frac{1}{2} * 2^{10} \text{ KB} = 2^{-1} * 2^{10} \text{ KB} = 2^9 \text{ KB}$
13. When we convert the number from one base to another base we only change the base and number remains same. So the odd number remains odd number and the even number remains as even number. In decimal numbers (base 10) even numbers are ends with 0, 2, 4, 6, 8 and odd numbers are ends with 1, 3, 5, 7, 9. When you change the even number to base 2 it ends with 0 and when you change the odd number it must ends with 1.
- ```

2 | 18
 | 0
2 | 9 1
 | 1
2 | 4 0
 | 0
2 | 2 0
 | 0
2 | 1 0
 | 0
 1

```
- $18 = 10010_2$  (18 is an even number so it must ends with 0)  
You may crosscheck the results by  $10010_2 = 1*16+0*8+0*4+1*2+0*1 = 18$
14. In order to find the decimal equivalent of binary number we need to write the numbers multiples of 2 starting with 1 from right to left.
- ```

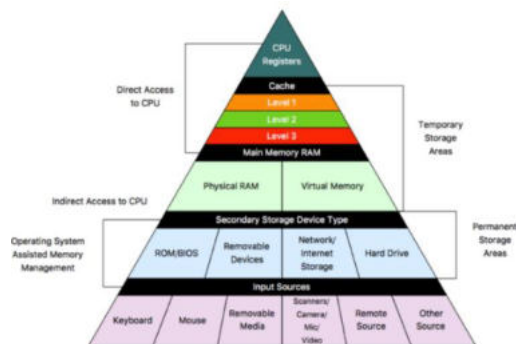
1      0      1      1
2^3    2^2    2^1    2^0
2*2*2  2*2  2      1
8        4      2      1

```
- $1011_2 = 8*1+4*0+2*1+1*1 = 8+4+1 = 11$
15. To convert the binary (base 2) numbers to octal (base 8) we should group the numbers from right to left as 3 digits ($8 = 2^3$) and convert it individually.
- e.g.
 101010_2
 $101|010_2$
 $5 | 2_8 = 52_8$
Convert the 10101_2 change the base to eight.
 $010|101_2$ (you can add an extra on the left to reduce the calculation mistakes).
16. To convert the binary (base 2) to hexadecimal (base 16) we should group the numbers from right to left as 4 digits ($16 = 2^4$). e.g. convert the binary number 10010010_2 to hexadecimal.
- ```

1001|0010_2
 9 | 2_{16} = 92_{16}

```
- e.g. convert the  $1010010_2$  to hexadecimal  
 $0101|0010_2$  (Adding zeros in front reduces the mistakes).  
 $5 | 2_{16} = 52_{16}$
17. Computer memory speeds in descending order
- Register (fastest memory type) inside the CPU.
  - Cache (level 1 cache is inside the CPU)
  - RAM (Random Access Memory)
  - SSD (Solid State Drive)
  - Hard Disk
  - Tape drive
18. Capacity of the memory types in ascending order
- Register - fastest memory and an expensive memory type.
  - Cache
  - CD (compact Disk - 700MB)
  - RAM (Usually from 2GB - 16GB)
  - DVD (Digital Versatile Disk - 4.7GB)
  - Blue Ray Disc (25GB)
  - SSD (Solid State Disk from 64GB - 1TB)
  - Harddisk (80GB - 4TB)

- i. Magnetic Tape (1TB - 25 TB). Unlike above memory types magnetic tapes uses sequential access.



19. **BCD** - Binary Coded Decimal means representing decimal numbers in binary format. We can represent digits from 0 to 9 in binary for that we need to use 4 bits as 9 in decimal equals to 1001 in binary. In BCD we cannot represent any other number like 10, 11, 12, 13, 14, 15 with 4 bits BCD and for that we should use 8bits (first 4 bits for 1 and rest for the balance number). We cannot represent letters in BCD.

$8_{10}$  in decimal equals to 1000 in BCD

$10_{10}$  (1|0) in decimal equals to 1|0000 in BCD

$4|9_{10} = 0100|1001$  BCD

$4|2_{10} = 0100|0010$  BCD

20. Encoding and the number of bits used

| Encoding                                                                                                                  | No of bits used  |
|---------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>BCD</b>                                                                                                                | 4                |
| <b>ASCII</b> (American Code for Information Interchange uses English Alphabets and numbers )                              | 7                |
| <b>EBCDIC</b> (Extended Binary Coded Decimal Interchange)                                                                 | 8                |
| <b>UNICODE</b> (Used to represent almost all the languages globally in a single encoding like Tamil, Sinhala and English) | 16 bits (2 Byte) |

21. Binary addition can be performed like decimal numbers.  $0_2+0_2=0_2$ ,  $0_2+1_2=1_2$ ,  $1_2+0_2=1_2$ ,  $1_2+1_2=10_2$ , (You may compare with  $0_{10}+0_{10}=0_{10}$ ,  $0_{10}+1_{10}=1_{10}$ ,  $1_{10}+0_{10}=1_{10}$ ,  $1_{10}+1_{10}=2_{10}$  and  $2_{10}=10_2$ ).

22. Binary subtractions can be performed like decimal numbers.

$100_2 - 0_2=100_2$ ,  $100_2 - 1_2 = 11_2$ ,  $100_2 - 10_2 = 10_2$ ,  $100_2 - 11_2 = 1_2$ ,  $100_2 - 1_2 = 11_2$

23. A is represented in ASCII as 100 0001<sub>2</sub>

a is represented in ASCII as 110 0001<sub>2</sub>

So in ASCII capital and small can be interchanged by changing a bit value.

24. In ASCII A ( 65<sub>10</sub>) is represented as 100 0001<sub>2</sub>.

B (65<sub>10</sub>+1<sub>10</sub>=66<sub>10</sub>) is represented as 100 0010<sub>2</sub> (100 0001<sub>2</sub>+1<sub>2</sub>=100 0010<sub>2</sub>

C(65<sub>10</sub>+2<sub>10</sub>=67<sub>10</sub>) is represented as 100 0010<sub>2</sub> (100 0001<sub>2</sub>+10<sub>2</sub>=100 0011<sub>2</sub>).

For example if A in ASCII refers to 1000001 in binary how D will be referred?

You may either refer B as A+1, C as A+2 and D as A+3 so D is 1000001<sub>2</sub>+11<sub>2</sub> (3<sub>10</sub>= 11<sub>2</sub>) = 1000100<sub>2</sub>

Alternatively as per ASCII A is 65, B is 66, C is 67, D is 68 you may convert 68 to binary as well

For example if E is referred as 1000101<sub>2</sub> in ASCII how B is encoded. Like in the previous example D = E-1, C = E-2, B = E-3

E - 1000101<sub>2</sub>

-3 11<sub>2</sub>

B - 1000010<sub>2</sub>

Like previous example you can also consider A as 65 and B as 66 then convert it to binary also results the same answer.

25. Ascending order, Descending order questions with the same base.

For example consider these three numbers 100<sub>2</sub>, 110<sub>2</sub>, 101<sub>2</sub>

We may either convert the given numbers in decimal i.e.

100<sub>2</sub> - 4

110<sub>2</sub> - 6

101<sub>2</sub> - 5

So if we arrange the numbers in ascending order numbers becomes

100<sub>2</sub> , 101<sub>2</sub>, 110<sub>2</sub>

If you want to do the calculation quickly when the given numbers are in the same

- base you can consider it as decimal numbers like  $100_2$ ,  $110_2$ ,  $101_2$  as one hundred, one hundred and ten, one hundred and one. Then you can easily write the numbers in ascending order as  $100_2$ ,  $101_2$ ,  $110_2$
26. For ascending order/descending order questions with different bases. First try to sort easiest ones to with base 8 or base 16 to same base like base 2

#### MCQ

- The smallest unit of storage is referred as a GIT 2005  
1)Mega Byte 2)Kilo Byte 3)Byte 4)Bit
- ASCII Stands for GIT 2005  
1)American Standard Code for Information Interchange  
2)American Standard Code for International Information  
3)American Statistical Code for International Information  
4)American Standard Code for Interchange Information
- The base of the hexadecimal number system is GIT 2006  
1)6 2)8 3)10 4)16
- Which of the following can be used as a coding scheme to represent characters? GIT2007  
1)EBCDIC 2)IEEE 3)ANSI 4)TURBO
- How many bytes are there in a kilobyte? OL2007MCQ13  
(1) 8 (2) 1000 (3) 1024 (4) 1048
- Which of the following statements are true? OL2007MCQ14  
A - A bit is the smallest unit of information that a computer can process.  
B - A bytes is the smallest unit of information that a computer can process.  
C - A bit contains 8 bytes.  
D - A bytes contains 8 bit  
(1) A and C (2) A and D (3) B and C (4) B and D
- The decimal equivalent  $1010_2$  is OL2007MCQ15  
1)5 2)6 3)10 4)12
- The binary equivalent of decimal 15 is OL2007MCQ16  
(1) 1001 (2) 1100 (3) 1111 (4) 10101
- Which of the following has the least value? OL2007MCQ17 (I only changed one answer below)  
1) $45_{10}$  2) $11000_2$  3) $37_8$  4) $1A_{16}$
- Consider the following statements regarding storage media. OL2008MCQ04  
(A) A typical floppy disk has less storage capacity than a compact disk  
(B) A modern hard disk has more storage capacity than a compact disk  
(C) A digital versatile disk has more storage capacity than a modern hard disk  
Which one of the following is true?  
(1) (A) only (2) (B) only (3) (A) and (B) only (4) (C) only
- In a Personal Computer (PC), an alphabetic, a numeric or a special character is represented by a ..... OL2008MCQ08  
(1) bit (2) nibble (3) byte (4) kilobyte
- The binary equivalent of the octal number  $36_8$  is OL2008MCQ09  
1) $110011_2$  2) $101110_2$  3) $11110_2$  4) $11111111_2$
- A megabyte contains: OL2008MCQ10  
1) $2^6$  Kilobytes 2) $2^{20}$  bytes 3) $10^6$  Bytes 4) $2^{20}$  bits
- The decimal equivalent of  $111000_2$  ..... OL2008MCQ11  
1) 7 2)38 3)56 4)70
- The binary equivalent of the decimal number 33 is OL2008MCQ12  
1)  $1111_2$  2) $011011_2$  3) $100010_2$  4) $100001_2$
- The value of  $A9_{16}$  is equivalent to ..... OL2008MCQ13  
1)  $169_{10}$  2) $10011010_2$  3) $1211_8$  4)Both  $169_{10}$  and  $1211_8$
- If  $P=$  and  $Q=$  then  $P+Q$  is ..... OL2008MCQ14  
1) $1111_2$  2) $E_{16}$  3) $F_{16}$  4)Both  $1111_2$  and  $F_{16}$

18. The binary equivalent of 67 is? GIT2008MCQ01  
1)1100001 2)1000011 3)1010001 4)1000101
19. The hexadecimal equivalent of the binary number 11011101010? GIT2008MCQ02  
1)6EA 2)DD2 3)3352 4)6722
20. What is the base of octal system? GIT2008MCQ03  
1)6 2)8 3)10 4)16
21. The sum of two binary numbers is 1010 and 11 is? GIT2008MCQ04  
1)1110 2)1011 3)1010 4)1101
22. The number of bits used in the UNICODE coding scheme is ? GIT2008MCQ08  
1)7 2)8 3)9 4)16
23. Which of the following storage media usually has the largest storage capacity?  
GIT2008MCQ36  
1)DVD 2)CD 3)Floppy Disk 4)Hard disk
24. The minimum number of bits required to represent the Hexa-decimal number  $ACE_{16}$  is  
OL2009MCQ08  
1)three 2)nine 3)eleven 4)twelve
25. What is the decimal equivalent of  $101011_2$  ? OL2009MCQ09  
1)27 2)41 3)43 4)51
26. 2 megabytes contain OL2009MCQ10  
1)  $2^7$  Bytes 2)  $2^{21}$  bytes 3)  $2 \times 10^6$  bytes 4)  $2 \times 10^{20}$  bytes
27. A student borrows Rs 50 from his friend and paid it back completely in two instalments A and B. Which of the following could be the equivalent of the instalments? OL2009MCQ11  
1)  $A=100010_2$  and  $B=1100_2$  2)  $A=100111_2$  and  $B=1010_2$  3)  $A=100011_2$  and  $B=1111_2$   
4)  $A=100001_2$  and  $B=1110_2$
28. The ASCII code of the character "A" is 65. Which one of the following represents the character "A" in binary? OL2009MCQ12  
1) 111111<sub>2</sub> 2)110101<sub>2</sub> 3)1000010<sub>2</sub> 4)1000001<sub>2</sub>
29. Which one of the following values is **not** equivalent to  $1AF_{16}$ ? OL2009MCQ13  
1)  $431_{10}$  2)110101111<sub>2</sub> 3)  $657_8$  4)  $567_8$
30. If  $P=1010_2$  and  $Q=1111_2$  then  $P+Q =$  OL2009MCQ14  
1) 11001<sub>2</sub> 2)  $1A_{16}$  3)  $AF_{16}$  4)10101<sub>2</sub>
31. The binary equivalent of the octal number 73 is? GIT2009MCQ01  
1)100101 2)111011 3)1001001 4)100111
32. The hexadecimal equivalent of the binary number 11010101110 is? GIT2009MCQ02  
1)6532 2)D56 3)3256 4)6AE
33. The base of the decimal number system is? GIT2009MCQ03  
1)2 2)8 3)10 4)16
34. The sum of two binary numbers 1001 and 101 is GIT2009MCQ04  
1)1110 2)1011 3)1010 4)1100
35. The number of bits used to represent a decimal digit in Binary Coded Decimal (BCD) coding scheme is GIT2009MCQ08  
1)2 2)4 3)6 4)8
36. Which of the following correctly arranges the access speed of the following in descending order? GIT2009MCQ36  
1)Cache Memory, RAM, DVD 2)Cache Memory, DVD, RAM  
3)DVD, Cache Memory, RAM 4)RAM, Cache Memory, DVD
37. 2GB of memory is equivalent to OL2010MCQ06  
1) 2048 KB 2)  $2^{11}$  MB 3)  $2^{10}$  MB 4)1024 MB
38. The octal equivalent of binary 11001 is? OL2010MCQ07  
1)21 2)31 3)37 4)41

39. Which of the following is the largest? OL2010MCQ08  
1) 1001100<sub>2</sub> 2) 1001010<sub>2</sub> 3) 1010101<sub>2</sub> 4) 1100000<sub>2</sub>
40. If the character 'C' represented as 1000011<sub>2</sub> in the ASCII code, what is the equivalent ASCII code for the character 'A'? OL2010MCQ09  
1) 1000001<sub>2</sub> 2) 1000010<sub>2</sub> 3) 1010101<sub>2</sub> 4) 1100000<sub>2</sub>
41. A teacher enters the number of students in the class as 35 to the computer using the keyboard. This number is represented in the Random Access Memory (RAM) of the computer in Binary Form. The number of students stored in the RAM is OL2010MCQ10  
1) 100010<sub>2</sub> 2) 100110<sub>2</sub> 3) 100111<sub>2</sub> 4) 100011<sub>2</sub>
42. The decimal equivalent of the binary number 1001101 is? GIT2010MCQ01  
1) 71 2) 77 3) 79 4) 83
43. The binary equivalent of the decimal number 137 is? GIT2010MCQ02  
1) 10001011 2) 1001001 3) 10001001 4) 1101001
44. Which of the following pairs of binary represents an Odd value and an Even value when converted into a decimal form? GIT2010MCQ04  
1) 0101, 0011 2) 0010, 1000 3) 0111, 0001 4) 0111, 0110
45. One GB (1GB) approximately equivalent to? GIT2010MCQ06  
1) 10KB 2) 10<sup>2</sup>KB 3) 10<sup>3</sup>KB 4) 10<sup>3</sup>MB
46. Age of father is equal to the sum of the ages of his daughter and his son. The ages of daughter and his son in binary form are 11110 and 11011 years respectively. What is the age of the father in decimal form? GIT2010MCQ07  
1) 55 2) 57 3) 59 4) 62
47. What is the equivalent value for decimal value 125? OL2011MCQ05  
1) 01011101<sub>2</sub> 2) 01111101<sub>2</sub> 3) 174<sub>8</sub> 4) 7B<sub>16</sub>
48. What value does the symbol 'C' represents in the hexadecimal number system? OL2011MCQ06  
1) 1011<sub>2</sub> 2) 1101<sub>2</sub> 3) 14<sub>8</sub> 4) 13<sub>10</sub>
49. Two computers, A and B have memory capacities of 1GB and 1024MB respectively. What is the correct statement regarding their memory capacities? OL2011MCQ07  
1) Computer A has the higher memory capacity than Computer B.  
2) Computer B has the higher memory capacity than Computer A.  
3) Both have equal memory capacities.  
4) Memory capacities of the two computers cannot be compared as different units of measurement are used.
50. Saman has an electronic presentation of size 10,256 bits. What is the minimum capacity of the portable flash drive that can be used for storing this file? OL2011MCQ08  
1) 256 Bytes 2) 1 KB 3) 1 MB 4) 1 GB
51. Binary Coded Decimal (BCD) uses four bits to represent each digit in the decimal number system. Which of the following is a valid bit pattern in BCD? OL2011MCQ09  
1) 0011 2) 1011 3) 1100 4) 1101
52. If 1000001<sub>2</sub> represents A in ASCII, which of the represented by 1000111<sub>2</sub>? OL2011MCQ10  
1) D 2) E 3) F 4) G
53. Parakrama receives Rs 40/= from his father and Rs 30/= from his mother to purchase a book. The binary equivalent of the total amount is? GIT2011MCQ01  
1) 1000011 2) 10000100 3) 1000101 4) 1000110
54. The main memory of the computer stores the year 2011 in binary form as? GIT2011MCQ02  
1) 11110011001 2) 11110011011 3) 11111011011 4) 11111111011
55. A flash memory device has a capacity of 4GB. Which of the following represents the capacity of the flash memory device in MB? GIT2011MCQ04  
1)  $\frac{1}{4} \times 10^3$  MB 2) 10<sup>3</sup> MB 3) 4 × 10<sup>3</sup> MB 5) 4 × 10<sup>6</sup> MB

56. Which of the following shows the storage devices arranged according to their access speeds from the slowest to the fastest? OL2012MCQ03  
 1) Floppy Disk, Hard Disk, Main Memory 2) Hard Disk, Floppy Disk, Main Memory  
 3) Hard Disk, Main Memory, Floppy Disk 4) Main Memory, Hard Disk, Floppy Disk
57. The binary equivalent of the number  $3A_{16}$  is ..... OL2012MCQ05  
 1) 111000 2) 111001 3) 111010 4) 111011
58. The decimal equivalent of the number  $1010101_2$  is OL2012MCQ06  
 1) 21 2) 58 3) 81 4) 85
59. Consider the number ' $11_x$ ' where 'x' is the base of the number system. Which of the following number system(s) can the number belong to? OL2012MCQ07  
 1) Binary only 2) Hexadecimal only 3) Octal Only 4) Either binary, Hexadecimal or Octal
60. Binary Coded Decimal (BCD) uses four bits to represent each digit in the decimal number system. What is the bit pattern used to encode the decimal number 28 in the BCD code? OL2012MCQ08  
 1) 00011100 2) 00101000 3) 10000010 4) 11100000
61. If ' $1000111_2$ ' represents 'G' in ASCII, what character does ASCII ' $1000101_2$ ' represent? OL2012MCQ09  
 1) A 2) B 3) C 4) D
62. The decimal equivalent to Binary  $10101101$  is? GIT2012MCQ01  
 1) 170 2) 173 3) 177 4) 191
63. The price of a pineapple is Rs.160/- and the price of a mango is Rs 40/-. The total price of pineapple and mango in Binary form is GIT2012MCQ02  
 1) 10001000 2) 11001000 3) 11011000 4) 11011010
64. A computer has a memory (RAM) capacity of 4GB. Which of the following is equivalent to the RAM capacity of the computer? OL2013MCQ04  
 1) 4096 Bytes 2) 4096 KB 3) 4096 MB 4) 4096 TB
65. Which of the following is equivalent to  $127_{10}$ ? OL2013MCQ09  
 1)  $011111101_2$  2)  $11111110_2$  3)  $177_8$  4)  $FF_{16}$
66. The binary equivalent of  $A9_{16}$  is OL2013MCQ10  
 1)  $10100100_2$  2)  $10101001_2$  3)  $10101100_2$  4)  $11101001_2$
67. Which one is the smallest out of the following, Binary Coded Decimal (BCD) representations? OL2013MCQ11  
 1) 01111001 2) 10000000 3) 10010101 4) 10011001
68. If the character 'a' is represented in ASCII as 1100001 then the character 'e' would be represented as: OL2013MCQ12  
 1) 1100001 2) 1100011 3) 1100100 4) 1100101
69. What is the binary equivalent of the decimal number 57? GIT2013MCQ01  
 1) 101101 2) 110001 3) 111011 4) 111001
70. Which of the following is equivalent to the decimal value  $131_{10}$ ? OL2014MCQ05  
 1)  $10110011_2$  2)  $10000011_2$  3)  $47_{16}$  4)  $76_8$
71. What is the decimal equivalent of binary  $101011_2$ ? OL2014MCQ06  
 1) 27 2) 35 3) 43 4) 75
72. Which of the following has the least value? OL2014MCQ07  
 1)  $11001001_2$  2)  $145_{10}$  3)  $77_8$  4)  $49_{16}$
73. Which of the following has the highest file size? OL2014MCQ08  
 1) .doc file of 1 Mega Byte 2) .gif file of 912 Kilo Bytes  
 3) .txt file of 700 Bytes 4) .jpg file of 1024 Bytes
74. If the character 'E' represented as 1000101 in ASCII encoding scheme, what is the ASCII code for the character 'B'? OL 2014MCQ09  
 1)  $1000001_2$  2)  $1000010_2$  3)  $1000011_2$  4)  $1100101_2$

75. Which of the following octal numbers is equivalent to decimal 79? OL2015MCQ05  
1) 117<sub>8</sub> 2) 1017<sub>8</sub> 3) 711<sub>8</sub> 4) 7110<sub>8</sub>
76. Which of the following formats are used in computers to store data? OL2015MCQ06  
1) Binary 2) Decimal 3) Hexa-decimal 4) Octal
77. Which of the following decimal numbers is equivalent to binary numbers 110111<sub>2</sub> ? OL2015MCQ07  
1) 55 2) 63 3) 110 4) 118
78. Which of the following represents the ascending order of the four numbers 10001111<sub>2</sub>, 170<sub>8</sub>, 46687<sub>10</sub>, 2F<sub>16</sub> ? OL2015MCQ08  
1) 170<sub>8</sub>, 10001111<sub>2</sub>, 2F<sub>16</sub>, 46687<sub>10</sub> 2) 10001111<sub>2</sub>, 46687<sub>10</sub>, 170<sub>8</sub>, 2F<sub>16</sub>  
3) 2F<sub>16</sub>, 170<sub>8</sub>, 10001111<sub>2</sub>, 46687<sub>10</sub> 4) 46687<sub>10</sub>, 10001111<sub>2</sub>, 170<sub>8</sub>, 2F<sub>16</sub>
79. Which of the file/files given in Table 1 could be stored in a USB storage device having a total capacity of 4GB so that a least amount of unused space is left on the device? OL2015MCQ09

Table 1

| File Name | Size  |
|-----------|-------|
| A.doc     | 300MB |
| B.jpg     | 740MB |
| C.mp4     | 3GB   |

- 1) A.doc and B.jpg only 2) B.jpg and C.mp4 only 3) C.mp4 only 4) All of them
80. Which of the following sentence(s) is/are true? GIT2015MCQ01  
A - The Hard disk has larger capacity than main memory.  
B - The data in main memory are lost when the power is turned off while the data in the hard disk are not.  
C - The data in main memory are stored in electronic circuits while the data in the hard disk are stored on magnetic media.  
1) A only 2) B only 3) C only 4) All A, B and C

81. Table lists four files and their sizes. GIT2015MCQ05  
Which of the files in the table could be stored in a flash memory drive having a total capacity of 4GB, if 3GB of the flash memory is already used?

| File | Size (MB) |
|------|-----------|
| A    | 224       |
| B    | 770       |
| C    | 815       |
| D    | 1025      |

- 1) A and B only 2) A and C only 3) B and C only 4) C and D only
82. Which of the following decimal number is equivalent to the binary number 100110? GIT2015MCQ06  
1) 27 2) 30 3) 38 4) 43
83. Which of the following binary numbers is equivalent to the decimal number 73? GIT2015MCQ07  
1) 1000101 2) 1001001 3) 1010001 4) 1100001
84. Which of the following is approximately equivalent to one Terabyte (TB)? OL2016MCQ04  
1) 1x10<sup>6</sup>MB 2) 1x10<sup>6</sup> GB 3) 1x10<sup>6</sup>KB 4) 1x10<sup>9</sup>Bytes
85. What is the decimal equivalent to octal 64? OL2016MCQ05  
1) 48 2) 52 3) 62 4) 68
86. Which of the following show data transfer speeds of secondary storage devices arranged in an ascending order? OL2016MCQ29  
1) Solid state hard drive, Magnetic Hard drive, floppy disk drive, DVD drive  
2) floppy disk drive, Magnetic hard drive, DVD drive, Solid state hard drive  
3) floppy disk drive, DVD drive, Magnetic hard drive, Solid state hard drive  
4) Solid state hard drive, Magnetic hard drive, DVD drive, floppy disk drive

87. In the ASCII code, Character 'Q' is represented as  $1010001_2$ . What is the option that represents the word 'PUT' in binary form? OL2016MCQ33  
 1)  $1010000_2$ ,  $1010101_2$ ,  $1010100_2$  2)  $1010001_2$ ,  $1010101_2$ ,  $1010100_2$   
 3)  $1010000_2$ ,  $1010011_2$ ,  $1010111_2$  4)  $1010101_2$ ,  $1010100_2$ ,  $1010011_2$
88. Consider the following four numbers presented in different number systems:  
 OL2016MCQ34  
 $11110011_2$ ,  $F3_{16}$ ,  $363_8$ ,  $243_{10}$   
 Which of the following statements is true regarding the above four numbers?  
 1)  $F3_{16}$  is larger than other three numbers  
 2)  $363_8$  is smaller than other three numbers  
 3) Four numbers are equal to each other  
 4) Four numbers are not equal to each other
89. Which of the following binary number is equivalent to the decimal number 35?  
 GIT2016MCQ01  
 1) 100001 2) 100011 3) 101011 4) 101010
90. Which of the following decimal numbers is equivalent to the binary number  $10001011_2$ ?  
 GIT2016MCQ10  
 1) 113 2) 139 3) 213 4) 231
91. The descending order of the three binary numbers  $1101101$ ,  $1101001$ ,  $1110010$  is  
 GIT2016MCQ31  
 1)  $1110010$ ,  $1101101$ ,  $1101001$  2)  $1101101$ ,  $1101001$ ,  $1110010$   
 3)  $1110010$ ,  $1101001$ ,  $1101101$  4)  $1101001$ ,  $1101101$ ,  $1110010$
92. In a certain coding system alphabetical characters A to Z are coded in the consecutive binary numbers. A is coded as  $1000000$ . What is the equivalent binary for character C? GIT2016MCQ39  
 1)  $1000001$  2)  $1000010$  3)  $1000011$  4)  $1000101$
93. Which of the following decimal numbers is equivalent to the binary number  $01011_2$ ?  
 OL2017MCQ01  
 1) 11 2) 35 3) 15 4) 10
94. Which of the following octal numbers is equivalent to the binary number  $111110110010_2$ ? OL2017MCQ02  
 1)  $7552_8$  2)  $2667_8$  3)  $2557_8$  4)  $7662_8$
95. The hexadecimal equivalent of  $11101011000111010_2$  is? OL2017MCQ03  
 1)  $DD63A_{16}$  2)  $1D63A_{16}$  3)  $1D33A_{16}$  4)  $1D631_{16}$
96. Consider the following statements: OL2017MCQ04  
 A -  $2B_{16}$  is equivalent to  $53_8$   
 B -  $43_{10}$  is equivalent to  $101011_2$   
 C -  $101011_2$  is equivalent to  $53_8$   
 Which of the above statements are true?  
 1) A and B only 2) A and C only 3) B and C only 4) All A, B and C
97. Consider the following statements OL2017MCQ10  
 A - Register memory in a computer has a smaller capacity than that of a hard disk.  
 B - Data access speed of register memory is slower than the speed of a hard disk  
 C - Per bit cost to store data in register memory is higher than per bit cost to store data in a hard disk  
 Which of the above statements are true?  
 1) A and B only 2) A and C only 3) B and C only 4) All A, B and C
98. Which of the following binary number is equivalent to the decimal number 40? GIT2017MCQ05  
 1)  $000100$  2)  $100000$  3)  $100100$  4)  $101000$
99. The descending order of three binary numbers  $01001101$ ,  $10110011$ ,  $11010011$  is  
 GIT2017MCQ06  
 1)  $01001101$ ,  $10110011$ ,  $11010011$  2)  $11010011$ ,  $10110011$ ,  $01001101$   
 3)  $11010011$ ,  $01001101$ ,  $10110011$  4)  $10110011$ ,  $11010011$ ,  $01001101$

100. Coding system is used to represent alphabetic and other characters in the binary system. If the alphabetic characters are represented in consecutive binary numbers in such a coding system, and if the letter M is coded as 1001101, then the letter O will be coded as? GIT2017MCQ07  
1) 1001011 2) 1001100 3) 1001110 4) 1001111
101. In which form is data and instructions stored in a computer? OL2018MCQ01  
1) binary 2) decimal 3) hexadecimal 4) octal
102. Which of the following is the octal equivalent of decimal  $156_{10}$ ? OL2018MCQ02  
1)  $121_8$  2)  $234_8$  3)  $574_8$  4)  $770_8$
103. Which of the following is the binary equivalent of hexadecimal  $2B_{16}$ ? OL2018MCQ03  
1) 00101011<sub>2</sub> 2) 01001001<sub>2</sub> 3) 10010100<sub>2</sub> 4) 10110011<sub>2</sub>
104. Which of the following lists  $10011001_2$ ,  $113_8$ ,  $160_{10}$ ,  $1A_{16}$  in the ascending order? OL2018MCQ04  
1)  $10011001_2$ ,  $113_8$ ,  $160_{10}$ ,  $1A_{16}$  2)  $160_{10}$ ,  $1A_{16}$ ,  $10011001_2$ ,  $113_8$   
3)  $160_{10}$ ,  $1A_{16}$ ,  $113_8$ ,  $10011001_2$  4)  $1A_{16}$ ,  $113_8$ ,  $10011001_2$ ,  $160_{10}$
105. In which of the following number systems, is the number '800' a valid number? OL2018MCQ05  
1) decimal only 2) decimal and hexadecimal only  
2) decimal and octal only 4) octal only
106. Which of the following statements are true? OL2019MCQ09  
A - Binary form is used to store data and instructions in computers  
B - 945 is a valid number both in octal and hexadecimal number systems  
C -  $412_8$  is equivalent to  $100001010_2$   
1) A only 2) B only 3) A and C only 4) All A, B and C
107. Which of the following shows the given storage components in descending order of access speed? OL2019MCQ10  
1) cache memory, main memory, register, hard disk  
2) hard disk, cache memory, register, main memory  
3) register, cache memory, main memory, hard disk  
4) register, main memory, hard disk, cache memory
108. If character 'E' is represented in the ASCII table as  $69_{10}$ , what is the binary representation of the character 'G' in the ASCII table? OL2019MCQ11  
1) 1000110 2) 1000111 3) 1001000 4) 1001001
109. Which of the following represents the result of the binary arithmetic operation  $11001100 - 01010101$ ? AL2019MCQ20  
1) 100110011 2) 01100110 3) 01110111 4) 10011001 5) 10101010
110. Which of the following represents the units of measurements of data in computer system in the ascending order of their size? OL2020MCQ03  
1) Bit, Byte, Kilobyte, Terabyte 2) Byte, Bit, Kilobyte, Terabyte  
3) Megabyte, Kilobyte, bit, Byte 4) Terabyte, Gigabyte, Megabyte, Kilobyte
111. Which of the following shows the storage devices of a desktop computer in descending order of storage capacity? OL2020MCQ04  
1) Register, Cache memory, Hard disk 2) Hard disk, RAM, Register  
3) RAM, Register, Cache memory 4) RAM, Cache memory, Hard disk
112. Consider the character representation in Table 1 and Table 2 given below AL2020MCQ08

Table 1:

|      |      |      |      |
|------|------|------|------|
| අ    | ආ    | ඇ    | ඈ    |
| 0D85 | 0D86 | 0D87 | 0D88 |

Table 2:

|      |      |      |      |
|------|------|------|------|
| அ    | ஆ    | இ    | ஈ    |
| 0B85 | 0B86 | 0B87 | 0B88 |

In which of the character encoding system given below, the above characters in Table 1 and Table 2 are represented?

- 1) Both Tables 1 and 2 : in ASCII  
2) Both Tables 1 and 2 : in UNICODE  
3) Table 1: in ASCII, Table 2: in UNICODE  
4) Table 1: in EBCDIC, Table 2: in ASCII  
5) Table 1: in UNICODE, Table 2: in ASCII

### Compulsory Questions

113. What is the minimum number of keys required to represent the English alphabet (A to Z, a to z) digits (0 to 9) and the ten special characters !, @, #, \$, %, ^, &, \* ( ) in a computer. OL 2009
114. Convert the binary number 11001110 into hexadecimal equivalent OL 2010
115. Covert the number of bytes in 1KB into octal equivalent OL 2010
116. Convert encodings schemes are used to represent symbols within computers. The number of symbols represented in a computer depends on the number of bits used for the encoding scheme. How many different symbols could be represented using eight (8) bits? Show your computation. OL 2011
117. Convert Hexadecimal number C9 in to binary. Show your computations. OL 2011
118. Convert the binary number 10101001 into hexadecimal. Show your computations. OL 2012
119. Name a coding system used to represent characters (e.g. @, 2, A, ....) in computers OL 2012
120. Convert the binary number 10101010 into decimal - OL 2013
121. Write down the main purpose of an encoding scheme such as ASCII - OL 2013
122. Convert the hexadecimal number A7B<sub>16</sub> into its equivalent binary number. Show your computations. OL 2014
123. Encoding schemes are used to represent characters and symbols with computers. List two different encoding schemes. OL 2014
124. Convert the binary 11010101<sub>2</sub> to the equivalent octal number. Show your steps. OL 2015
125. In the ASCII code, the letter A is represented as decimal 65. Determine the binary representation of the letter D. OL 2015
126. Covert the Hexadecimal number E9 to the equivalent octal number. Show the relevant steps. OL 2016
127. a)Covert the binary number 110111001100<sub>2</sub> to octal. Show steps in your calculation OL 2017  
b)Convert the hexadecimal number 752<sub>2</sub> to binary. Show setps in your calculation.

### Answers

1. Byte
2. American Standard Code for Information Interchange
3. 16
4. EBCDIC
5. 1024
6. A and D
7. 10
8. 1111
9. 11000<sub>2</sub>
10. (A) and (B) only
11. Byte
12. 11110<sub>2</sub>
13. 2<sup>20</sup> bytes
14. 56
15. 100001<sub>2</sub>
16. 169<sub>10</sub>
17. Both 1111<sub>2</sub> and F<sub>16</sub>
18. 1000011
19. 6EA
20. 8
21. 1101
22. 16
23. Hard disk
24. 12 (There are three hexadecimal numbers and each hexa decimal number denoted by 4 bits so 3\*4=12 bits)
25. 43
26. 2<sup>21</sup> bytes
27. A=100011<sub>2</sub>, B=1111<sub>2</sub>
28. 1000001<sub>2</sub>
29. 567<sub>8</sub>
30. 11001<sub>2</sub>
31. 73<sub>8</sub> = 111011<sub>2</sub>
32. 6AE
33. 10
34. 1110
35. BCD uses 4bits (2nd Answer)
36. Cache Memory, RAM, DVD
37. 2GB = 2x2<sup>10</sup>MB = 2<sup>11</sup> MB
38. 11001<sub>2</sub> = 11|001<sub>2</sub> = 31<sub>8</sub>

39. As the number of digits are same for all the answers compare it without considering the base 2 so the answer is  $1100000_2$
40.  $B = C-1$ ,  $A = C-2$  so we need to deduct 2 i.e. in binary 10 so the answer is  $1000001_2$ , alternatively if you can remember the ASCII chart 65 stands for A so the answer in Binary is  $1000001_2$
41.  $32+2+1=35$  so =  $100011_2$
42.  $1001101_2 = 64+8+4+1=77$
43.  $128+8+1=137 = 10001001_2$
44. Odd number ends with one and even number ends with zero so the answer is 0111, 0110
45.  $10^3\text{MB}$
46. Either you can add it in binary format i.e.  $11110_2 + 11011_2 = 111001_2 = 32+16+8+1=57$   
Alternatively you can convert the given numbers to decimal and add them.
47.  $01111101_2$  (It's easy to do the subtraction  $127 - 2 = 01111111_2 - 10_2 = 01111101_2$ )
48.  $C_{16} = 12_{10} = 1100_2 = 1|100_2 = 14_8$
49. Both have equal memory capacities
50. To convert the Bit to Byte we need to divide it by 8.  $10,256\text{Bit}/8=128$  Byte from the given answers 1MB is the most suitable answer.
51. BCD - Binary Coded Decimal i.e. to write the decimal numbers in Binary format the decimal number needs to be in the range 0 -9. So the most suitable answer is 0011.
52.  $1000111_2 - 1000001_2 = 110_2 = 6$   
 $A, B = A+1, C = A+2, D = A+3, E = A+4, F = A + 5, G = A+6$  so the answer is G.
53.  $40+30=70=64+4+2=1000110_2$
54.  $2011 = 2047-36=2047-32-4=11111011011_2$
55.  $4 \times 10^3 \text{ MB}$  ( $4\text{GB} = 4 \times 1\text{GB}$ ,  $1\text{GB} \approx 10^3 \text{ MB}$  so  $4\text{GB} \approx 4 \times 10^3 \text{ MB}$ )
56. Floppy Disk, Hard Disk, Main Memory
57.  $3A_{16} = 3|A_{16} = 3|10_{10} = 0011|1010_2 = 111010_2$
58.  $1010101_2 = 64+16+4+1=85$
59. Either binary, Hexadecimal or Octal
60. To convert to BCD we have to convert the decimal numbers individually  $28 = 2|8 = 0010|1000 = 00101000$
61.  $1000111_2 - 1000010_2 = 101_2 = 5$  Write the alphabets in the order A, B, C, D, E, F, G then comes back in reverse direction (F = G-1, E = G-2, D = G-3, C = G-4, B = G-5 so the answer is B). Otherwise from the ASCII chart A =  $1000001_2$ , B =  $1000010_2$  (i.e. B = A+1).
62.  $10101101_2 = 128+32+8+4+1 = 173$
63.  $160+40=200=128+64+8=11001000_2$
64.  $4\text{GB} = 4 \times 1024\text{MB} = 4096\text{MB}$
65.  $177_8 = 7+8 \times 7+64=127$
66.  $A9_{16} = A|9_{16} = 10|9_{10} = 1010|1001_2 = 10101001_2$
67.  $01111001$  (look for zero in front of the numbers for smallest number)
68.  $a = 1100001$ ,  $b = a+1$ ,  $c = a+2$ ,  $d = a+3$ ,  $e = a+4$ ,  $a = 1100001_2$ ,  $4 = 100_2$ ,  $e = 1100101_2$
69.  $57 = 111001_2$
70.  $131 = 128+2+1 = 10000011_2$
71.  $101011_2 = 32+8+2+1=43$
72. Compare with base 2  
 $11001001_2 = 11|001|001_2 = 311_8$  so obviously  $77_8$  is smaller than  $11001001_2$ .  $49_{16} = 9+64=73$  and  $77_8 = 56+7=63$  so the smallest one is  $77_8$
73. .doc file of 1 Mega Byte
74. Method 1 Write the letters in the alphabetical order A, B, C, D, E. As per ASCII D=E-1, C=E-2, B=E-3.  $3 = 11_2$  E-3= $1000101_2 - 11_2 = 1000010_2$   
Alternatively knowing the ASCII Chart A (i.e. 65) =  $1000001_2$  and B =  $A+1 = 1000001_2 + 1_2 = 1000010_2$
75. We can easily eliminate the even number  $7110_8$  as the question is about the odd number.  
$$\begin{array}{r} 8 \overline{) 79} \\ 8 \overline{) 9} \underline{-7} \\ 1-1 \end{array}$$
  
Writing from bottom to top  $117_8$
76. Binary
77. We can easily eliminate the two even answers as the questions is about the odd number.  $110111_2 = 32+16+4+2+1=55$
78. Given numbers are in base 2, 8, 10, 16,. It's easy to convert the binary to octal and hexadecimal number. First lets convert the hexadecimal to binary  $2F_{16} = 0010|1111_2$  so  $2F_{16}$  is smaller than  $10001111_2$ . Questions is about the ascending order so after  $2F_{16}$  only  $10001111_2$  should come. There is only one answer.  $2F_{16}$ ,  $170_8$ ,  $10001111_2$ ,  $46687_{10}$
79. B.jpg and C.mp4 only
80. All A, B and C
81. A and B only
82. We can easily eliminate the two odd number answers.  $100110_2 = 32+4+2=38$
83.  $73=64+8+1=1001001_2$

84.  $1\text{TB} \approx 1000\text{GB} \approx 10^3 \text{ GB}$   
 $1\text{GB} \approx 1000\text{MB} \approx 10^3 \text{ MB}$   
 $1\text{TB} \approx 1000 \times 1000\text{MB} \approx 10^3 \times 10^3 \text{ MB} \approx 10^6 \text{ MB}$
85.  $64_8 = 8 \times 6 + 4 = 48 + 4 = 52$
86. floppy disk drive, DVD drive, Magnetic hard drive, Solid state hard drive
87. PQRSTU so  $P = Q-1$ ,  $U = Q+4$ ,  $T = Q+3$  The given number is an odd number so PUT should be even, odd, even. There is only one such answer namely  $101000_2$ ,  $101010_2$ ,  $1010100_2$
88.  $11110011_2 = 1111|0011_2 = F3_{16}$   
 $11110011_2 = 11|110|011_2 = 363_8$   
 Out of the given answers only suitable one is four numbers are equal to each other.
89.  $35 = 32 + 2 + 1 = 100011_2$
90.  $10001011_2 = 128 + 8 + 2 + 1 = 139$
91. Compare them without considering it as binary numbers.  $1110010$ ,  $1101101$ ,  $1101001$
92. ABC so  $B = A+1$ ,  $C = A+2 = 1000000_2 + 10_2 = 1000010_2$
93.  $01011_2 = 8 + 2 + 1 = 11$
94.  $11110110010_2 = 111|110|110|010_2 = 7662_8$
95.  $1|1101|0110|0011|1010_2 = 1D33A_{16}$
96.  $2B_{16} = 2|B_{16} = 10|1011_2 = 101|011_2 = 53_8$   
 $43_{10} = 32 + 8 + 2 + 1 = 101011_2$   
 All A, B and C
97. A and C only
98.  $40 = 32 + 8 = 101000_2$
99.  $11010011$ ,  $10110011$ ,  $01001101$  Consider these numbers as decimal numbers instead of binary.
100. MNO so  $O = M+2$   $O = 1001101_2 + 10_2 = 1001111_2$
101. Binary - Computer stores data in binary format and displays that in hexadecimal format (decimal format is human-readable as we are more familiar with decimal numbers from the childhood. Octal was used in the past with early computers and not now).
102. 
$$\begin{array}{r} 8 \overline{) 156} \\ 8 \overline{) 19} - 4 \\ \hline 2 - 3 \end{array}$$
  
 $= 234_8$  (Also note that the given number is an even number as the last digit 6 is even so the converted number will also be even so it cannot be  $121_8$  as the last digit 1 is an odd number).
103.  $2B_{16} = 2|B_{16}$ , A-  $10$ , B -  $11$   

$$\begin{array}{r} 2 \overline{) 11} \\ 2 \overline{) 5} - 1 \\ 2 \overline{) 2} - 1 \\ \hline 1 - 0 \end{array}$$
  
 $B_{16} = 1011_2$   
 From the last step  $2_{16} = 10_2$  as the given number is in hexadecimal ( $16 = 2^4$  we need to make sure that the length of the converted digit is 4 in order to do so we add two zeros in front) =  $0010_2$   
 Substituting for  $2B_{16} = 2_{16}|B_{16} = 00101011$  (We may able to drive the answer only be converting  $2_{16}$  to binary from the given answers).
104. Either we can convert the whole thing to a specific base say for example base 2 or will look at the given digits we look at two and three-digit numbers like  $1A_{16} = 1 \times 16^1 + A \times 16^0 = 16 + 10 \times 1 = 26_{10}$  compare with the  $160_{10}$ ,  $1A_{16} < 160_{10}$  from the given answers  $1A_{16}$ ,  $113_8$ ,  $10011001_2$ ,  $160_{10}$ .  
 The second approach is to convert all the digits to a common base we select base 2.  
 $1A_{16}$  A=ten in hexadecimal  

$$\begin{array}{r} 2 \overline{) 10} \\ 2 \overline{) 5} - 0 \\ 2 \overline{) 2} - 1 \\ \hline 1 - 0 \end{array}$$
  
 $1A_{16} = 00011010_2$   
 $113_8 = 1_8 | 1_8 | 3_8$   
 $= 001_2 | 001_2 | 011_2$  ( $8 = 2^3$  so needs three digits 1 should be written as 001)  
 $= 001001011_2$   

$$\begin{array}{r} 2 \overline{) 160} \\ 2 \overline{) 80} - 0 \\ 2 \overline{) 40} - 0 \\ 2 \overline{) 20} - 0 \\ 2 \overline{) 10} - 0 \\ 2 \overline{) 5} - 0 \\ 2 \overline{) 2} - 1 \\ \hline 1 - 0 \end{array}$$
  
 $= 10100000_2$   
 To arrange numbers in ascending order minimum length to maximum length disregarding zeros in front then look for the 1 with the highest power at the end. ( $2^4 < 2^5$ )  
 $11010_2$ ,  $1001011_2$ ,  $10011001_2$ ,  $10100000_2$

- Alternatively, we can convert all the numbers to decimal as reach the solution.
105. decimal and hexadecimal only (decimal numbers from 0-9 and hexadecimal numbers 0-15 (written as 0-9,A,B,C, D, E, F) and octal numbers from 0-7)
  106. A and C only (Decimal numbers from 0-9 similarly octal numbers from 0-7 so statement B is incorrect)
  107. register, cache memory, main memory, hard disk
  108. 1000111 ( $E = 69_{10}$  From the alphabetic order  $F = 70_{10}$  and  $G = 71_{10}$ )
  109. 11001100<sub>2</sub>  
 - 01010101<sub>2</sub>  
 =01110111<sub>2</sub> (When borrow from the previous digit remember  $2^1=2$  as the base is 2)
  110. Bit, Byte, Kilobyte, Terabyte (Byte=8bit, 1KB=1024Byte, 1TB = 1024\*1024KB).  
 Ascending order means arrange the items from the lower value to a high value.
  111. Hard disk, RAM, Register (Hard disk generally in Terabyte like 1TB, RAM in Gigabyte like 8GB, Register in KB).
  112. Both Tables 1 and 2 are in UNICODE